



SIMATS ENGINEERING



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ASSESSMENT TOOL -1

CO3_AT1

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EXPERIMENTS

1.Design a simple LAN consisting of two PCs connected through a switch using Cisco Packet Tracer. Configure appropriate IP addresses for both hosts and verify connectivity using the ping command. Simultaneously capture the packet exchange using Wireshark. Identify the Ethernet frame fields such as Source MAC Address, Destination MAC Address, EtherType, and Frame Check Sequence (FCS). Explain how the Data Link layer uses MAC addressing to ensure successful frame delivery within the local network.

ANSWER :-

LAN Using Switch – Ethernet Frame Analysis

Aim

To design a simple LAN with two PCs connected through a switch, configure IP addresses, verify connectivity, and analyze Ethernet frames using Wireshark.

Topology

PC0 ----- Switch ----- PC1

IP Configuration

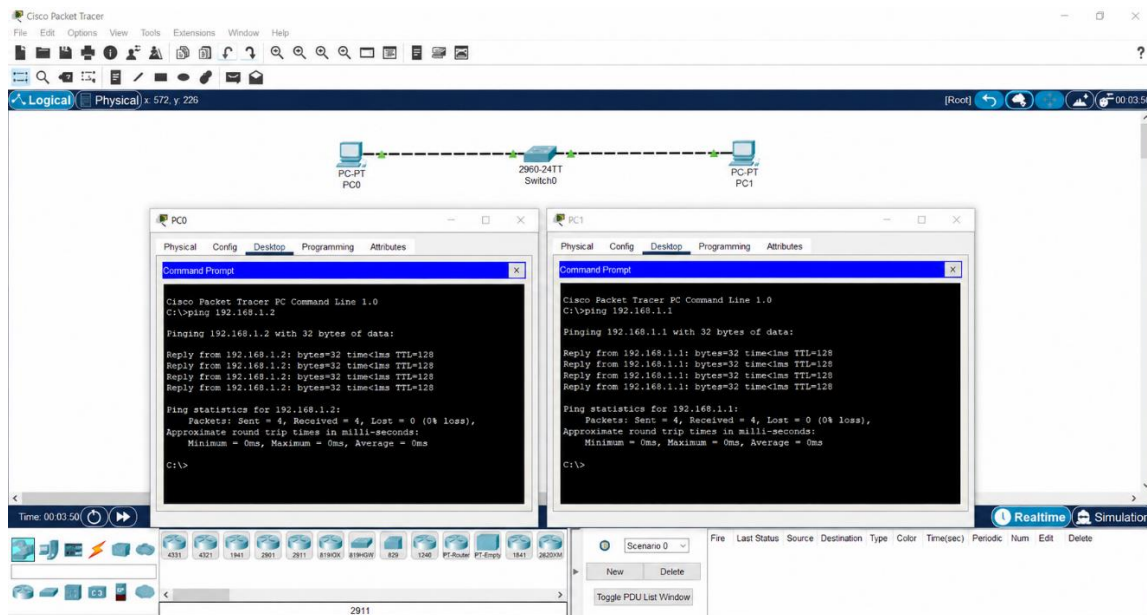
Device	IP Address	Subnet Mask
PC0	192.168.1.1	255.255.255.0
PC1	192.168.1.2	255.255.255.0

Procedure

- Open Cisco Packet Tracer.
- Place one switch and two PCs.
- Connect using Copper Straight-Through cables.
- Configure IP addresses on both PCs.
- Open Command Prompt on PC0.

- Execute:
- ping 192.168.1.2
- Capture packets using Wireshark.
- Stop capture after successful ping.

OUTPUT :-



Observation

Ethernet Frame contains:

- Destination MAC Address: Receiver's MAC address
- Source MAC Address: Sender's MAC address
- EtherType: Indicates upper-layer protocol (IPv4 = 0x0800)
- Payload: Actual IP packet
- Frame Check Sequence (FCS): Detects transmission errors

Result

Ping was successful with 0% packet loss.

2. Create a network topology consisting of two LANs connected through a router in Cisco Packet Tracer. Configure IPv4 addressing, subnet masks, and default gateways for all devices, and verify end-to-end communication using the ping command. Capture the ICMP packets using Wireshark and analyze the IPv4 header fields, including Source IP Address, Destination IP Address, Time-To- Live (TTL), Protocol Number, and Header Length. Explain the role of the Network layer in forwarding packets between different networks.

ANSWER:-

Two LANs Connected Through Router – IPv4 Analysis

Aim

To configure two LANs connected by a router and analyze IPv4 packets.

Topology

PC0 ---- Switch ---- Router ---- Switch ---- PC1

IP Configuration

Device	Interface	IP Address
PC0	NIC	192.168.1.2
Router	G0/0	192.168.1.1
Router	G0/1	192.168.2.1
PC1	NIC	192.168.2.2

Default Gateway:

- PC0 → 192.168.1.1
- PC1 → 192.168.2.1

Procedure

- Build topology.
- Configure router interfaces.
- Assign IP addresses.

- Set default gateways.
- Ping from PC0:
- ping 192.168.2.2
- Capture packets in Wireshark.

IPv4 Header Analysis

- Source IP: 192.168.1.2
- Destination IP: 192.168.2.2
- TTL: Typically 128 or 64 (decreases by 1 at router)
- Protocol Number: 1 (ICMP)
- Header Length: 20 bytes

OUTPUT :-

The screenshot displays the Cisco Packet Tracer interface with a network topology and two command-line windows.

Network Topology: A central 2811 Router0 is connected to two 2960-24TT switches. The left switch is connected to PC0 (192.168.1.2) and the right switch to PC1 (192.168.2.2). The router interfaces are labeled G0/0 (192.168.1.1) and G0/1 (192.168.2.1).

PC0 Command Prompt:

```
C:\>ping 192.168.2.2
Pinging 192.168.2.2 with 32 bytes of data:
Reply from 192.168.2.2: bytes=32 time=1ms TTL=126
Reply from 192.168.2.2: bytes=32 time=1ms TTL=126
Reply from 192.168.2.2: bytes=32 time=1ms TTL=126
Reply from 192.168.2.2: bytes=32 time=1ms TTL=126
Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 2ms, Average = 1ms
C:\>
```

PC1 Command Prompt:

```
C:\>ping 192.168.1.2
Pinging 192.168.1.2 with 32 bytes of data:
Reply from 192.168.2.2: bytes=32 time=1ms TTL=126
Reply from 192.168.2.2: bytes=32 time=1ms TTL=126
Reply from 192.168.2.2: bytes=32 time=1ms TTL=126
Reply from 192.168.2.2: bytes=32 time=1ms TTL=126
Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 2ms, Average = 1ms
C:\>
```

Simulation Panel: Shows an event list with timestamps and device names for ICMP Echo Request and Echo Reply messages between PC0, Switch0, Router0, Switch1, and PC1.

OSI Model Inbound PDU Details:

Field	Value
Internet Protocol Version 4	Version: 4
0100 ...	Header Length: 20 Bytes (5)
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)	
Total Length: 80	
Identification: 0x0001 (1)	
Flags: 0x00	
Source IP Address	192.168.1.2
Destination IP Address	192.168.2.2
Time to Live (TTL)	126
Protocol	1 (ICMP)
Header Length	20 Bytes

Result

Successful communication between different networks.

Explanation

The Network Layer (Layer 3) forwards packets between different networks using IP addresses. Routers examine the destination IP, consult their routing table, decrement TTL, and forward packets toward the destination network.

3. Construct a local area network with multiple hosts connected through a switch in Cisco Packet Tracer. Clear the ARP cache on one of the hosts and initiate communication with another host using the ping command. Capture the communication in Wireshark and identify the ARP Request and ARP Reply packets. Analyze the sender and target MAC/IP addresses and explain how the Address Resolution Protocol resolves logical IP addresses into physical MAC addresses before data transmission.

ANSWER :-

ARP Protocol Analysis

Aim

To observe ARP Request and ARP Reply packets using Wireshark.

Topology

PC0 ----- Switch ----- PC1

IP Configuration

Device	IP Address
PC0	192.168.1.1
PC1	192.168.1.2

Procedure

➤ Configure the LAN.

➤ Clear ARP cache:

arp -d

➤ Start Wireshark capture.

Ping:

➤ ping 192.168.1.2

- Observe ARP packets.

ARP Request

- Sender IP: 192.168.1.1
- Sender MAC: PC0 MAC
- Target IP: 192.168.1.2
- Target MAC: 00:00:00:00:00:00

Message:

- Who has 192.168.1.2?
- Tell 192.168.1.1

ARP Reply

- Sender IP: 192.168.1.2
- Sender MAC: PC1 MAC
- Target IP: 192.168.1.1
- Target MAC: PC0 MAC

Message:

192.168.1.2 is at AA:BB:CC:DD:EE:FF

OUTPUT:-

The screenshot displays the Cisco Packet Tracer interface. The network topology consists of two PCs, PC0 (192.168.1.1) and PC1 (192.168.1.2), connected via a 2960-24TT switch. The Command Prompt on PC0 shows the following commands and output:

```

C:\>arp -d
Deleting ARP entries ...

C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:
Reply from 192.168.1.2: bytes=32 time=1ms TTL=128
Reply from 192.168.1.2: bytes=32 time=1ms TTL=128
Reply from 192.168.1.2: bytes=32 time=1ms TTL=128
Reply from 192.168.1.2: bytes=32 time=1ms TTL=128

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>
  
```

The Simulation Panel shows the Event List with the following entries:

Vis.	Time(sec)	Last Device	At Device	Type	Info
	0.000	PC0	PC0	ARP	ARP Request
	0.001	Switch0	PC1	ARP	ARP Request
	0.002	PC1	Switch0	ARP	ARP Reply
	0.003	Switch0	PC0	ARP	ARP Reply
	0.004	PC0	PC1	ICMP	Echo Request
	0.005	PC1	PC0	ICMP	Echo Reply
	0.006	PC0	PC1	ICMP	Echo Request
	0.007	PC1	PC0	ICMP	Echo Reply

The Event List Filters - Visible Events section shows the following events:

ACL Filter, ARP, BGP, Bluetooth, CAPWAP, CDP, DHCP, DHCPv6, DNS, DTP, EAPOL, EIGRP, EIGRPv6, FTP, H.323, HSRP, HTTP, HTTPS, ICMP, ICMPv6, IPsec, ISAKMP, IoT, IoT TCP, LACP, LLDP, Meraki, NDP, NETFLOW, NTP, OSPF, OSPFv6, PAgP, PCP3, RADIUS, RIP, RIPv6, RTP, SCCP, SMTP, SNMP, SSH, STP, SYSLOG, TACACS, TCP, TFTP, Telnet, UDP, USB, VTP

Result

ARP successfully resolved the destination MAC address.

4. Develop a simple network consisting of two PCs connected through a router in Cisco Packet Tracer. Configure the required IP addresses and test connectivity using the ping command. Capture the ICMP traffic using Wireshark and identify the Echo Request and Echo Reply messages. Examine important ICMP fields such as Type, Code, Identifier, and Sequence Number, and calculate the Round Trip Time (RTT). Discuss how ICMP assists in network troubleshooting and connectivity verification.

ANSWER :-

Aim

To analyze ICMP Echo Request and Echo Reply packets.

Topology

PC0 ---- Router ---- PC1

IP Configuration

Device	IP Address
PC0	192.168.1.2
Router G0/0	192.168.1.1
Router G0/1	192.168.2.1
PC1	192.168.2.2

Procedure

- Configure router and PCs.
- Start Wireshark.
- Execute:

ping 192.168.2.2

- Observe ICMP packets.
- ICMP Fields

- Type 8: Echo Request
 - Type 0: Echo Reply
 - Code: 0
 - Identifier: Unique session ID
 - Sequence Number: Packet sequence number

Round Trip Time (RTT)

RTT is calculated as:

RTT = Reply Time – Request Time

Example:

Request = 2.450 ms

Reply = 3.260 ms

RTT = 0.810 ms

OUTPUT :-

The screenshot displays the Cisco Packet Tracer interface. At the top, a network topology is shown with the following components and connections:

- PC0** (192.168.1.2) connected to **Switch0** (2960-24TT) via G0/0 (192.168.1.1).
- Switch0** connected to **Router0** (2811) via G0/1 (192.168.2.1).
- Router0** connected to **Switch1** (2960-24TT) via G0/1 (192.168.2.1).
- Switch1** connected to **PC1** (192.168.2.2).

Below the topology, two command-line windows are open:

- PC0 Command Prompt:**

```
C:\>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:

Reply from 192.168.2.2: bytes=32 time<1ms TTL=125
Reply from 192.168.2.2: bytes=32 time<1ms TTL=125
Reply from 192.168.2.2: bytes=32 time<1ms TTL=125
Reply from 192.168.2.2: bytes=32 time<1ms TTL=125

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 2ms, Average = 1ms

C:\>
```
- PC1 Command Prompt:**

```
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time<1ms TTL=125
Reply from 192.168.1.2: bytes=32 time<1ms TTL=125
Reply from 192.168.1.2: bytes=32 time<1ms TTL=125
Reply from 192.168.1.2: bytes=32 time<1ms TTL=125

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 2ms, Average = 1ms

C:\>
```

On the right side, the **PDU Information at Device: Router0** window is open, showing the following details:

OSI Model	
Inbound PDU Details	
Source IP Address	: 192.168.1.2
Destination IP Address	: 192.168.2.2
Protocol	: ICMP (1)
Type	: 8 (Echo Request)
Code	: 0
Identifier	: 1
Sequence Number	: 1
Time To Live (TTL)	: 125
Header Length	: 20 Bytes

Below the table, it states: "This is an Echo Request ICMP packet."

Result

ICMP packets successfully exchanged.

5. Create a client-server network in Cisco Packet Tracer by configuring a web server and a client PC. Enable the HTTP service on the server and access the hosted webpage from the client using a web browser. Capture the communication in Wireshark and identify the TCP three-way handshake, followed by the HTTP GET request and HTTP response messages. Analyze the application-layer communication and explain how TCP provides reliable delivery for HTTP traffic through connection establishment, acknowledgments, and ordered data transfer.

ANSWER :-

HTTP Client–Server Communication

Aim

To configure a web server and client, access a webpage, and analyze TCP and HTTP communication.

Topology

Client PC ----- Switch ----- Web Server

IP Configuration

Device	IP Address
Client	192.168.1.2
Server	192.168.1.10

Subnet Mask:

255.255.255.0

Procedure

- Connect client and server through a switch.
- Configure IP addresses.
- Enable HTTP service on the server.

- Open browser on client.
- Enter:
- http://192.168.1.10
- Capture packets using Wireshark.
- TCP Three-Way Handshake
- SYN → Client requests connection.
- SYN-ACK → Server acknowledges.
- ACK → Client confirms connection.
- Connection established.
- HTTP Communication
- Client sends:
- HTTP GET /
- Server responds:
- HTTP/1.1 200 OK
- followed by the webpage contents.

Analysis

- TCP ensures reliable communication through:
- Connection establishment (Three-Way Handshake)
- Sequence Numbers
- Acknowledgments (ACK)
- Retransmission of lost packets
- Ordered delivery
- Flow control

HTTP uses TCP (port 80) to transfer web pages reliably.

OUTPUT :-

The screenshot displays two windows from a network simulation. The top window is Cisco Packet Tracer, showing a network topology with a PC (192.168.1.2) connected to a switch (2960-24TT) and a web server (192.168.1.1). The 'Web Server' configuration window is open, showing the 'Services' tab with HTTP enabled. The 'PC0' window shows a web browser displaying the 'Packet Tracer Web Server' page.

The bottom window is Wireshark, showing a packet capture on the 'tcp.stream == 0' filter. The packet list shows the following packets:

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	192.168.1.2	192.168.1.1	TCP	74	53014 → 80 [SYN] Seq=0 Win=8192 Len=0
2	0.000207	192.168.1.1	192.168.1.2	TCP	74	80 → 53014 [SYN, ACK] Seq=0 Ack=1 Win=8192 Len=0
3	0.000318	192.168.1.2	192.168.1.1	TCP	66	53014 → 80 [ACK] Seq=1 Ack=1 Win=64240 Len=0
4	0.001126	192.168.1.2	192.168.1.1	HTTP	184	GET / HTTP/1.1
5	0.001369	192.168.1.1	192.168.1.2	TCP	66	80 → 53014 [ACK] Seq=1 Ack=119 Win=65152 Len=0
6	0.001924	192.168.1.1	192.168.1.2	HTTP	396	HTTP/1.1 200 OK (text/html)
7	0.002177	192.168.1.2	192.168.1.1	TCP	66	53014 → 80 [ACK] Seq=119 Ack=331 Win=64240 Len=0

The packets are grouped into three categories:

- TCP Three-Way Handshake:** Packets 1, 2, and 3.
- HTTP GET Request:** Packet 4.
- HTTP Response:** Packet 6.

Result

The client successfully accessed the webpage hosted on the server, and Wireshark captured the TCP handshake, HTTP GET request, and HTTP response.

Conclusion

The experiment demonstrates application-layer communication using HTTP over TCP. TCP guarantees reliable, ordered, and error-free delivery, while HTTP enables web page retrieval between the client and server.